From Particle Systems to the Landau Equation

Xuanrui Feng

Beijing International Center for Mathematical Research
Peking University

April 3, 2026
Sino-Russian Student Mathematical Seminar

Landau Equation

Consider the space-homogeneous Landau equation in the integral-differential form

$$\partial_t f = \int_{\mathbb{R}^3} (\nabla_v - \nabla_w) \cdot \Big(a(v-w) \cdot (\nabla_v - \nabla_w) f(v) f(w) \Big) \, \mathrm{d}w,$$

or in the more commonly used divergence form

$$\partial_t f = \nabla \cdot [(a * f) \nabla f - (b * f) f].$$

Here $a(z) = |z|^\gamma (|z|^2 \mathrm{Id} - z \otimes z) \geq 0$ and $b(z) = \nabla \cdot a(z) = -2|z|^\gamma z$.

The unknown function $f(t, v)$ represents the velocity distribution in a plasma.

The parameter $\gamma \in [-3, 1]$ represents the regularity of the interaction forces. Note that $\gamma = -3$ corresponds to the most important **Coulomb interaction**.

One usually speaks of hard potentials if $\gamma \in (0, 1]$, Maxwellian molecules if $\gamma = 0$, moderately soft potentials if $\gamma \in [-2, 0)$, and very soft potentials if $\gamma \in [-3, -2)$.

Wellposedness: Villani 1998, Desvillettes–Villani 2000a,b, Silvestre 2017, Guillen–Silvestre 2025.

History of the Landau Equation

Basic Question: How can we describe a complicated system of a vast number of identical interacting particles, such as a rarefied gas or a plasma?

Different scales and governing principles:

Microscopic scale: state vector of all particles $(x_1, v_1, \cdots, x_N, v_N) \in (\mathbb{R}^3 \times \mathbb{R}^3)^N$ governed by Newton's second law of motion. Difficult to track and analyze mathematically since $N \sim 10^{23}$.

Macroscopic scale: spatial observables such as the mass $\rho(t, x)$, the velocity $u(t, x)$ and the temperature $T(t, x)$ governed by fluid equations. Losing information about the velocity distribution.

Mesoscopic scale: one-particle density function $f(t, x, v)$ governed by kinetic equations. Intermediate between the above two scales.

Foundings of the Kinetic Theory

Maxwell (1867) and **Boltzmann** (1872): Boltzmann equation and Maxwellian equilibrium for a rarefied gas undergoing short-range potentials.

$$\partial_t f = \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} \Big(f(v')f(w') - f(v)f(w) \Big) B(|v-w|, \cos\theta)\, \mathrm{d}\sigma\, \mathrm{d}w.$$

Here $v' = \frac{v+w}{2} + \frac{|v-w|}{2}\sigma$ and $w' = \frac{v+w}{2} - \frac{|v-w|}{2}\sigma$, and $\cos\theta = \sigma \cdot \frac{v-w}{|v-w|}$.

The interaction kernel B, depending on the physical modeling of gas molecules, has certain singularity. For a plasma where particles interact with Coulomb potentials, the Boltzmann equation does not make sense.

Landau (1936): Landau equation as an alternative to the Boltzmann equation when the charged particles perform Coulomb collisions in a plasma, by assuming that **the grazing collisions prevail**.

Question: Can we derive the Landau equation from certain many-particle systems?

Kac's Model

Kac (1956): derivation of kinetic equations from many-particle systems.

Initial state space: N i.i.d. velocities $V_1, \cdots, V_N$ with initial law f_0 in $\mathbb{R}^3$.
Collision rate: for each pair of indices (i, j), choose a random collision time T_{ij} according to some exponential distribution according to the interaction kernel. Let $T = \min T_{ij}$ be the collision time and change the corresponding pair of velocities.
Post-collisional state space:

$$V_i' = \frac{V_i + V_j}{2} + \frac{|V_i - V_j|}{2}\sigma, \quad V_j' = \frac{V_i + V_j}{2} - \frac{|V_i - V_j|}{2}\sigma,$$

where $\sigma \in \mathbb{S}^2$ is drawn according to the interaction kernel.
Kac's model for Landau: passing to the limit of grazing collisions as in the formal derivation of the Landau equation from the Boltzmann equation.

Landau Master Equation

Statistical approach: The joint law of Kac's model $F_N(t, v_1, \cdots, v_N)$ is governed by the Landau master equation:

$$\partial_t F_N = \frac{1}{N} \sum_{i<j}^{N} (\nabla_{v_i} - \nabla_{v_j}) \cdot \Big(a(v_i - v_j) \cdot (\nabla_{v_i} - \nabla_{v_j}) F_N \Big).$$

The joint law F_N is symmetric since the particles are indistinguishable.
The statistical information is contained in the marginals $F_{N,k}$ of F_N:

$$F_{N,k}(t, v_1, \cdots, v_k) = \int_{\mathbb{R}^{3(N-k)}} F_N(t, v_1, \cdots, v_N) \, \mathrm{d}v_{k+1} \cdots \mathrm{d}v_N,$$

for any fixed k.
The sequence of probability measures $\{F_{N,k}\}$ solves the BBGKY hierarchy: The evolution equation of $F_{N,k}$ involves $F_{N,k+1}$.

Formal Derivation assuming Kac's Chaos

We write down the equation solved by the first marginal $F_{N,1}$:

$$\partial_t F_{N,1} = \frac{N-1}{N} \nabla \cdot \int_{\mathbb{R}^3} a(v-w) \cdot (\nabla_v - \nabla_w) F_{N,2} \, \mathrm{d}w.$$

If we assume that $F_{N,2}(t,v,w) = F_{N,1}(t,v) F_{N,1}(t,w)$ for any $t \geq 0$, then we derive the Landau equation by passing $N \to \infty$.

But for any fixed N, we cannot expect $F_{N,2}(t,v,w) = F_{N,1}(t,v) F_{N,1}(t,w)$, even if we start from i.i.d. initial data $F_N(0) = F_{N,1}(0)^{\otimes N}$, since the particles interact with each other and produce correlations.

Relaxation: The notion of **Kac's chaos**. Although independence is annihilated, it is reproduced after taking the limit $N \to \infty$.

Propagation of Chaos

Definition 1 (Kac's chaos)

Let E be a Polish space. Let f be a probability measure on E and let F_N be a symmetric probability measure on E^N. The sequence of measure $(F_N)_{N\geq 1}$ is called f-chaotic, if for any fixed k, we have $F_{N,k} \rightharpoonup f^{\otimes k}$ as $N \to \infty$.

Kac's chaos represents some kind of "asymptotic independence".

Definition 2 (Propagation of (Kac's) chaos)

Kac's chaos is said to be propagated if the following diagram commutes.

$$\begin{array}{ccc} F_{N,k}(0) & \rightharpoonup & f^{\otimes k}(0) \\ \Downarrow_{\text{particle}} & & \Downarrow_{\text{kinetic}} \\ F_{N,k}(t) & \rightharpoonup & f^{\otimes k}(t) \end{array} \cdot$$

In other words, if $F_N(0)$ is $f(0)$-chaotic, then for any $t \geq 0$, we have that $F_N(t)$ is $f(t)$-chaotic, which means Kac's chaos is propagated to any positive time.

Kac's Program for the Landau Equation

Theorem (F.–Wang (2025+))

Let $f_0 \in L^1 \cap L^\infty(\mathbb{R}^3)$ be a probability density. Let $F_N \in L^\infty([0,T]; L^1 \cap L^\infty(\mathbb{R}^{3N}))$ be the unique bounded weak solution to the Landau master equation with initial data $f_0^{\otimes N}$. Let $f \in C^1((0,\infty); \mathcal{S}(\mathbb{R}^3)) \cap L^\infty([0,\infty); L^1 \cap L^\infty(\mathbb{R}^3))$ be the unique bounded smooth solution to the Landau equation with initial data f_0.
Assume that f_0 has finite weighted Fisher information and high-order L^1 moment:

$$\int_{\mathbb{R}^3} \langle v \rangle^3 |\nabla \log f_0|^2 f_0 \, \mathrm{d}v < \infty, \quad \int_{\mathbb{R}^3} |v|^m f_0 \, \mathrm{d}v < \infty \text{ for some } m > 6.$$

Then for any $T > 0$, we have propagation of chaos: for any $k \geq 1$, the k-marginal $F_{N,k}$ converges weakly to $f^{\otimes k}$ as $N \to \infty$ on $[0,T] \times \mathbb{R}^{3k}$:

$$\int_{\mathbb{R}^{3k}} (F_{N,k} - f^{\otimes k}) \varphi^{\otimes k} \, \mathrm{d}v_1 \cdots \mathrm{d}v_k \to 0$$

as $N \to \infty$, for any $\varphi \in C_c^\infty(\mathbb{R}^3)$ and any $t \in [0,T]$. Here $\langle v \rangle = \sqrt{1 + |v|^2}$.

Related Works

Kac's program: McKean 1967, Carlen–Carvalho–Le Roux–Loss–Villani 2010, Mischler–Mouhot 2013, Hauray–Mischler 2014, Carrapatoso 2015, Fournier–Mischler 2025+.

Landau master equation: Balescu–Prigogine 1959, Kiessling–Lancelloti 2004, Miot–Pulvirenti–Saffirio 2011.

Kac's program for Landau: Carrapatoso 2016 (for Maxwellian molecules), Fournier–Guillin 2017 (for hard potentials).

Particle approximation for Landau: Fontbona–Guérin–Méléard 2009, Fournier 2010, Fournier–Hauray 2016, Carrillo–F.–Guo–Jabin–Wang 2026, Du–Li 2024+.

Limitation: all particle approximation results miss the case of very soft potentials. Kac's model has not been justified even for all soft potentials.

Coincidence: Tabary uploaded his preprint on arXiv solving the same problem exactly one day later. A truncation is needed on the particle system.

Duality Formulation

We adapt the duality formulation initiated by Bresch–Duerinckx–Jabin for McKean–Vlasov equations. Consider the backward master equation

$$\partial_t \Phi_N = -\frac{1}{N}\sum_{i<j}^{N}(\nabla_{v_i} - \nabla_{v_j}) \cdot \Big(a(v_i - v_j) \cdot (\nabla_{v_i} - \nabla_{v_j})\Phi_N\Big),$$

with final data $\Phi_N(T,\cdot) = (C_N^k)^{-1}\sum_{i_1<\cdots<i_k}\varphi(v_{i_1})\cdots\varphi(v_{i_k})$.

Motivation: Instead of comparing two **time-dependent measures** with a **fixed observable**, we construct a **time-dependent observable** according to one measure and use it to **observe the other measure**. Φ_N corresponds to the observable along the particle trajectory.
This idea is more intrinsic for kinetic equations with various conservation laws. A similar idea has been considered in Mischler–Mouhot 2013.

By direct computation,

$$\int_{\mathbb{R}^{3k}} F_{N,k}(T)\varphi^{\otimes k} - \int_{\mathbb{R}^{3k}} f(T)^{\otimes k}\varphi^{\otimes k} = \int_{\mathbb{R}^{3N}} F_N(T)\Phi_N(T) - \int_{\mathbb{R}^{3N}} f(T)^{\otimes N}\Phi_N(T)$$

$$= \int_{\mathbb{R}^{3N}} F_N(0)\Phi_N(0) - \int_{\mathbb{R}^{3N}} f(T)^{\otimes N}\Phi_N(T) = -\int_0^T \int_{\mathbb{R}^{3N}} \frac{\mathrm{d}}{\mathrm{d}t} f(t)^{\otimes N}\Phi_N(t).$$

It suffices to prove the asymptotic vanishing of the integral

$$\int_0^T \int_{\mathbb{R}^{3N}} \frac{\mathrm{d}}{\mathrm{d}t} f(t)^{\otimes N}\Phi_N(t).$$

Heuristics: the evolution of a *fixed* observable along the two trajectories asymptotically coincide $\Longrightarrow$ the observable *along the particle trajectory* asymptotically remains unchanged *along the Landau trajectory*.

By direct computation, this is equal to (up to a small error)

$$N \int_0^T \int_{\mathbb{R}^{3N}} V_f(v_1, v_2) f^{\otimes N} \Phi_N.$$

Here V_f is a test function with a specific form

$$V_f(v_1, v_2) = \phi(v_1, v_2) - \mathbb{E}_f \phi(v_1, \cdot) - \mathbb{E}_f \phi(\cdot, v_2),$$

where $\phi(v, w)$ is equal to

$$\frac{1}{f(v)f(w)} (\nabla_v - \nabla_w) \cdot \Big(a(v - w) \cdot (\nabla_v - \nabla_w) f(v) f(w) \Big).$$

Obviously V_f satisfies a two-side zero-expectation condition.

$$\mathbb{E}_f V_f(\cdot, w) = 0, \quad \mathbb{E}_f V_f(v, \cdot) = 0.$$

Cluster Expansion

We apply the classical cluster expansion to reduce the correlation quantity to the binary correlation.

$$\Phi_N(v_1, \cdots, v_N) = \sum_{n=0}^{N} \sum_{\sigma \in P_n^N} C_{N,n}(v_\sigma)$$

Here P_n^N represents the set of all subsets of $\{1, \cdots, N\}$ with exactly n elements. $v_\sigma = \{v_i : i \in \sigma\}$.
The correlation function $C_{N,n}$ satisfies the zero-expectation condition:

$$\int_{\mathbb{R}^3} C_{N,n}(v_1, \cdots, v_n) f(v_i)\, \mathrm{d}v_i = 0, \forall 1 \leq i \leq n.$$

Inversion formula:

$$C_{N,n}(v_1, \cdots, v_n) = \sum_{m=0}^{n} (-1)^{n-m} \sum_{\sigma \in P_m^n} \int_{\mathbb{R}^{3(N-m)}} \Phi_N f^{\otimes(N-m)}\, \mathrm{d}v_{\sigma^c}.$$

Applying the cluster expansion formula, we have

$$N\int_0^T\int_{\mathbb{R}^{3N}} V_f(v_1,v_2)f^{\otimes N}\Phi_N = \int_0^T\iint_{\mathbb{R}^6} V_f(v,w)f(v)f(w)NC_{N,2}(v,w)\,\mathrm{d}v\,\mathrm{d}w\,\mathrm{d}t.$$

Therefore, it suffices to prove the following two results:

- $V_f \in L^1([0,T];L^2(f^{\otimes 2}\,\mathrm{d}v\,\mathrm{d}w))$;
- $NC_{N,2} \rightharpoonup^* 0$ in $L^1([0,T];L^2(f^{\otimes 2}\,\mathrm{d}v\,\mathrm{d}w))$.

The regularity estimate of V_f requires a new commutator inequality and a second-order Fisher information estimate to deal with the Coulomb singularity.

The weak convergence of $NC_{N,2}$ is obtained by a compactness argument and an energy method for deriving the uniqueness of the limit hierarchy.

Estimating V_f by Second-Order Fisher Information

- V_f contains both:
  - the **Coulomb singularity**;
  - unbounded Fokker–Planck quantities $\nabla \log f$, $\nabla^2 \log f$.
- We focus on controlling the key double integral

$$\iint_{\mathbb{R}^6} \frac{|\nabla \log f(v) - \nabla \log f(w)|^2}{|v-w|^4} f(v) f(w) \, \mathrm{d}v \, \mathrm{d}w.$$

- Coulomb singularity: $|v-w|^{-4}$ is not locally integrable, so we need to go beyond the local L^2 assumptions of Bresch–Duerinckx–Jabin.
- We also lack a $W^{1,\infty}$ bound on $\nabla \log f$, so we cannot directly integrate by parts or apply a simple mean-value argument.

Fractional Sobolev Structure and Fisher Information

- Main commutator-type estimate (schematically):

$$|\nabla \log f(v) - \nabla \log f(w)|^2 f(v) f(w) \lesssim \left(|\nabla \sqrt{f}(v)|^2 - |\nabla \sqrt{f}(w)|^2\right)^2$$

$$+ |f(v) - f(w)|^2 + |\nabla f(v) - \nabla f(w)|^2.$$

- Plugging this into the double integral and using the representation of the homogeneous fractional Sobolev norm

$$\|g\|^2_{\dot{H}^{1/2}(\mathbb{R}^3)} \sim \iint_{\mathbb{R}^6} \frac{|g(v) - g(w)|^2}{|v - w|^4} \, \mathrm{d}v \, \mathrm{d}w,$$

  we reduce the problem to Sobolev norms of $\sqrt{f}$.
- This leads to bounds in terms of a **second-order Fisher information**, essentially controlling $\|\sqrt{f}\|_{H^2}$.
- Remaining step: propagating a weighted Fisher information along the Landau flow, using its diffusive structure to control the time integral of the second-order Fisher information on $[0, T]$.

Landau Master Equation and Maximum Principle

- The Landau master equation has a special divergence form.
- First key observation: the dual solution Φ_N coincides with the time reverse of a (weak) solution of the master equation, with initial data given by the terminal data of Φ_N.
- We prove a parabolic maximum principle:

$$\|F_N(t)\|_{L^\infty(\mathbb{R}^{3N})} \leq \|F_N(0)\|_{L^\infty(\mathbb{R}^{3N})}, \quad t \geq 0.$$

- Consequently, we obtain a uniform L^∞ bound on Φ_N:

$$\|\Phi_N\|_{L^\infty((-\infty,T];L^\infty(\mathbb{R}^{3N}))} \leq \|\varphi\|^k_{L^\infty(\mathbb{R}^3)}.$$

- This L^∞ control is a cornerstone for the subsequent hierarchy estimates.

Correlation Hierarchy and Weak-$*$ Limits

- From the L^∞ bound on Φ_N and orthogonality of $C_{N,n}$, we obtain an *a priori* Hilbert norm estimate:

$$\sup_N \|N^{n/2} C_{N,n}\|_{\mathcal{H}} < \infty,$$

  where $\mathcal{H} = L^2(f^{\otimes n})$ is a weighted L^2 space.

- By Banach–Alaoglu, for each fixed n we can extract a weak convergent subsequence such that $N^{n/2} C_{N,n} \rightharpoonup^* \bar{C}_n$.

- It remains to identify the limits $\bar{C}_n$ and to show $\bar{C}_2 = 0$.

- To this end, we
  - write the finite-N hierarchy satisfied by $C_{N,n}$;
  - pass to the weak-$*$ limit along a convergent subsequence;
  - obtain a limiting hierarchy for $\bar{C}_n$ and justify the uniqueness.

Advantages of the Dual Hierarchy

- Compared to the classical BBGKY hierarchy, the limit hierarchy for $\bar{C}_n$ enjoys two key advantages:

1. **No higher-order derivatives:** derivatives of higher-order correlation functions vanish as $N \to \infty$, so the limit hierarchy does not involve uncontrolled higher-order terms.

2. **Balanced degeneracy:** although the energy dissipation term is degenerate, the other derivative terms degenerate at the same rate, allowing us to control them via the dissipation using Cauchy–Schwarz.

- Consequence: uniqueness for the limit hierarchy follows by a standard energy method, which forces $\bar{C}_2 = 0$ and yields the desired propagation of Kac's chaos.

The Landau–Maxwellian equation

Now we consider the case with Maxwellian molecules, i.e. when $\gamma = 0$.
Recall the Landau equation in the following form:

$$\partial_t f = \nabla \cdot [(a * f)\nabla f - (b * f)f], \quad \text{or} \quad \partial_t f = (a * f) : \nabla^2 f - (c * f)f,$$

with the vector field $b = \nabla \cdot a$ and the scalar $c = \nabla \cdot b$, namely

$$b_\alpha(v) = \partial_\beta a_{\alpha\beta}(v) = -(d-1)v_\alpha, \quad c(v) = \partial_\alpha b_\alpha(v) = -d(d-1).$$

The Landau system conserves the total mass, momentum and kinetic energy of the gas. We normalize the quantities to

$$\int_{\mathbb{R}^d} f(v)\,\mathrm{d}v = 1, \quad \int_{\mathbb{R}^d} vf(v)\,\mathrm{d}v = 0, \quad \int_{\mathbb{R}^d} |v|^2 f(v)\,\mathrm{d}v = d.$$

Particle Systems for the Landau–Maxwellian

We are interested in deriving the Landau equation as the mean-field limit of many particle system. We consider the Nanbu particle system with N indistinguishable particles, in the sense of a probabilistic interpretation:

$$\mathrm{d}V_t^i = \frac{2}{N}\sum_{j=1}^{N} b(V_t^i - V_t^j)\,\mathrm{d}t + \sqrt{2}\Big(\frac{1}{N}\sum_{j=1}^{N} a(V_t^i - V_t^j)\Big)^{\frac{1}{2}}\,\mathrm{d}B_t^i, \quad i = 1, \cdots, N.$$

We use the convention that $a(0) = 0$ and $b(0) = 0$ to omit the notation $i \neq j$. Applying Itô's formula and the relation $\nabla \cdot a = b$, we derive the master equation of the N-particle joint law $F_N(t, v^1, \cdots, v^N)$ on $\mathbb{R}^{dN}$:

$$\partial_t F_N = \sum_{i=1}^{N} \mathrm{div}_{v_i}\Big[\frac{1}{N}\sum_{j=1}^{N} a(v^i - v^j)\nabla_{v^i} F_N - \frac{1}{N}\sum_{j=1}^{N} b(v^i - v^j) F_N\Big],$$

where the initial value is fully tensorized as $F_N(0) = f_0^{\otimes N}$.

Entropic Propagation of Chaos

Theorem (Carrillo–F.–Guo–Jabin–Wang (2026 JMPA))

For any $T > 0$, we assume that classical solution $f \in \mathcal{C}^1([0,T], \mathcal{C}^2(\mathbb{R}^d))$ solves the Landau equation with $f \geq 0$ and the conservation laws. Assume that the initial data $f_0 \in \mathcal{C}^2(\mathbb{R}^d)$ and there exists some positive constants C_1, C_2 and C_3 such that f_0 satisfies the Gaussian-type bound from above that

$$f_0(v) \leq C_1 \exp(-C_1^{-1}|v|^2),$$

and the logarithm growth conditions

$$|\nabla \log f_0(v)| \leq C_2(1+|v|), \quad |\nabla^2 \log f_0(v)| \leq C_3(1+|v|^2).$$

Then for any entropy solution of the Liouville equation, we have the following estimate

$$\mathcal{H}_N(F_N|f_N)(t) \leq \mathcal{H}_N(F_N|f_N)(0) + \frac{C_T}{\sqrt{N}}.$$

Relative Entropy

We use the (scaled) relative entropy to quantify chaos

$$0 \leq \mathcal{H}_N(F_N|f^{\otimes N})(t) = \frac{1}{N}\int_{E^N} F_N \log \frac{F_N}{f^{\otimes N}}\,\mathrm{d}x_1 \cdots \mathrm{d}x_N.$$

Thanks to the monotonicity of the (scaled) relative entropy

$$\mathcal{H}_k(F_{N,k}|f^{\otimes k}) := \frac{1}{k}\int_{E^N} F_{N,k} \log \frac{F_{N,k}}{f^{\otimes k}}\,\mathrm{d}x_1 \cdots \mathrm{d}x_k \leq \mathcal{H}_N(F_N|f^{\otimes N})$$

and the classical Csiszár-Kullback-Pinsker inequality

$$\|F_{N,k} - f^{\otimes k}\|_{L^1} \leq \sqrt{2k\mathcal{H}_k(F_{N,k}|f^{\otimes k})},$$

one can obtain *propagation of chaos* given a vanishing sequence of $H_N(F_N|f^{\otimes N})$.

Relative entropy for propagation of chaos: Jabin–Wang 2016 JFA, 2018 Inventiones, Guillin–Le Bris–Monmarché 2024 JEMS, F.–Wang 2026 PMJ.

Main Ideas

We calculate the time derivative of the relative entropy as

$$\frac{\mathrm{d}}{\mathrm{d}t} H_N(F_N|f_N)(t) \leq -\frac{1}{N}\sum_{i=1}^{N}\int_{\mathbb{R}^{dN}} F_N \frac{1}{N}\sum_{j=1}^{N} a(v^i - v^j) : \nabla_{v^i} \log \frac{F_N}{f_N} \otimes \nabla_{v^i} \log \frac{F_N}{f_N}\,\mathrm{d}V$$
$$+ \frac{1}{N}\sum_{i=1}^{N}\int_{\mathbb{R}^{dN}} F_N \Big[a * f(v^i) - \frac{1}{N}\sum_{j=1}^{N} a(v^i - v^j)\Big] : \frac{\nabla^2_{v^i} f_N}{f_N}\,\mathrm{d}V.$$

By Cauchy–Schwarz inequality, the right-hand side can be further bounded by

$$\Big(\int_{\mathbb{R}^{dN}} F_N \frac{1}{N}\sum_{i=1}^{N}\Big|a * f(v^i) - \frac{1}{N}\sum_{j=1}^{N} a(v^i - v^j)\Big|^2\Big)^{\frac{1}{2}} \Big(\int_{\mathbb{R}^{dN}} F_N \frac{1}{N}\sum_{i=1}^{N}\Big|\frac{\nabla^2 f}{f}(v^i)\Big|^2\Big)^{\frac{1}{2}}.$$

We can then conclude combining estimates on $\nabla \log f$ and $\nabla^2 \log f$ and a new Law of Large Numbers based on the conservation laws.

Proposition 1 (Law of Large Numbers)

Assume that $T \sim O(N)$, we have

$$\frac{1}{N}\sum_{i=1}^{N}\int_{\mathbb{R}^{dN}} F_N \left| a * f(v^i) - \frac{1}{N}\sum_{j=1}^{N} a(v^i - v^j)\right|^2 \mathrm{d}V \lesssim \frac{1+T}{N}.$$

Using the explicit polynomial form of the test function and carefully computing the evolution of the moments of F_N at all orders.

Proposition 2 (Logarithmic Growth Estimates)

Assume that $f \in \mathcal{C}^1([0,T], \mathcal{C}^2(\mathbb{R}^d))$ with its initial data satisfying the same conditions as in the main result. Then for any $t \in [0,T]$ we have

$$|\nabla \log f| \lesssim 1 + \sqrt{t} + |v|, \quad |\nabla^2 \log f| \lesssim 1 + t + |v|^2.$$

Using Hamilton-type heat kernel estimates by the parabolic maximum principle.

Further Discussion

1. Quantitative Kac's program for the Landau equation; Convergence in better sense or uniform-in-time estimates.
2. Similar results for the Boltzmann equation or the Vlasov–Poisson equation.
3. Central limit theorem/ Gaussian fluctuation/ Large deviation principles for kinetic equations.
4. Derivation of spatially inhomogeneous kinetic equations from Newton dynamics under low-density limit or weak-coupling limit.

References

- X. Feng and Z. Wang, Kac's program for the Landau equation. arXiv preprint arXiv:2506.14309 (2025).
- J. Carrillo, X. Feng, S. Guo, P.-E. Jabin and Z. Wang, Relative entropy method for particle approximation of the Landau equation for Maxwellian molecules. **J. Math. Pures Appl. 206** (2026) 103838.
- S. Mischler and C. Mouhot, Kac's program in kinetic theory. **Invent. Math. 193** (2013) 1-147.
- D. Bresch, M. Duerinckx and P.-E. Jabin, A duality method for mean-field limits with singular interactions. arXiv preprint arXiv:2402.04695 (2024).
- P.-E. Jabin and Z. Wang, Quantitative estimates of propagation of chaos for stochastic systems with $W^{-1,\infty}$ kernels. **Invent. Math. 214** (2018) 523-591.
- X. Feng and Z. Wang, Quantitative propagation of chaos for 2D viscous vortex model on the whole space. **Peking Math. J.** (2026).

Thank you!